kader Academy

Machine Safety Specialist (CE)

Tab 1 – Module 1: Legislation on CE marking

TAB1: principles machine safety
Tab1-2/5 20231010

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Tab1-1/5-Sh.2

Arbeidsongevallen in Nederland

Monitor Arbeidsongevallen 2020

Nederlandse Arbeidsinspectie
Ministerie van Sociale Zaken en Werkgelegenheid

Monitor arbeidsongevallen 2021

Monitor Arbeidsongevallen 2020	Monitor Arbeidsongevallen 2021
Gemelde arbeidsongevallen: 3655	Gemelde arbeidsongevallen: 3482
Dodelijke arbeidsongevallen: 54	Dodelijke arbeidsongevallen: 62

- De belangrijkste ongevalstypen zijn al een aantal jaren hetzelfde, namelijk vallen en in contact komen met een arbeidsmiddel.
- Slachtoffers zijn vaak man en vallen vaak in de jongste (jonger dan 24) en de oudere leeftijdsgroepen (45 en ouder).

2

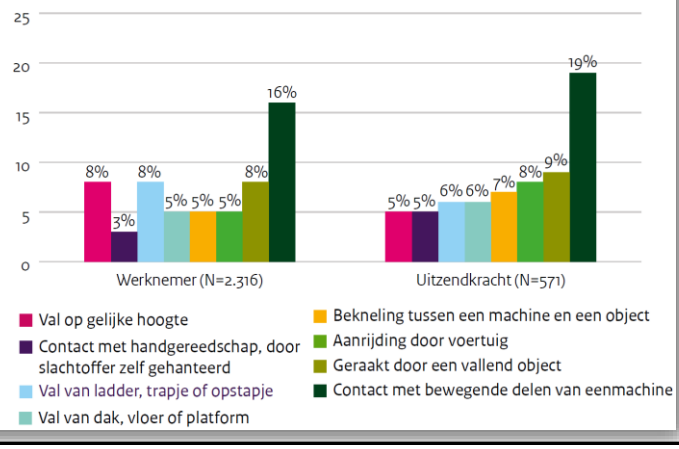
Tab1-1/5-Sh.3

Arbeidsongevallen in Nederland

Nederlandse Arbeidsinspectie
Ministerie van Sociale Zaken en Werkgelegenheid

Monitor arbeidsongevallen
2021

Figuur 3.5 Ongevallen met uitzendkrachten en werknemers, naar ongevalstype over 2020-2021[28]



Tab1-1/5-Sh.4

Arbeidsongevallen in Nederland

Inspectie SZW: betere preventie van arbeidsongevallen is noodzakelijk

Nieuwsbericht | 28-07-2021 | 10:00

Bedrijven kunnen nog steeds veel verbeteren waar het gaat om het voorkomen van arbeidsongevallen. Dat concludeert de Inspectie SZW in de Monitor arbeidsongevallen 2020 die vandaag is gepubliceerd. Vorig jaar zijn er 3655 arbeidsongevallen bij de Inspectie SZW gemeld. Er overleden 54 mensen als gevolg van een ongeval tijdens het werk. Door de verschillende lockdowns en het thuiswerken is het aantal ongevallen lager dan voorgaande jaren.

Nederlandse Arbeidsinspectie
Ministerie van Sociale Zaken en Werkgelegenheid

Bij de ongevallen met een machine was meestal sprake van contact met bewegende delen van een machine. Dit heeft vaak ernstige gevolgen. Bij een aanzienlijk deel van de slachtoffers is sprake van blijvend letsel, zoals amputatie van vingers of handen. Slachtoffers van machineongevallen waren in ongeveer de helft van de gevallen de machine aan het bedienen, de andere helft was bezig met schoonmaak of onderhoud. In dat laatste geval stond de machine vaak nog aan of standby, terwijl het slachtoffer dacht dat de machine uit stond. Veel voorkomende oorzaken van deze ongevallen zijn het betreden van de gevarenzone van de machine of mankementen aan de fysieke afscherming. In bijna de helft van de ongevallen was sprake van een combinatie van beide.

Introduction

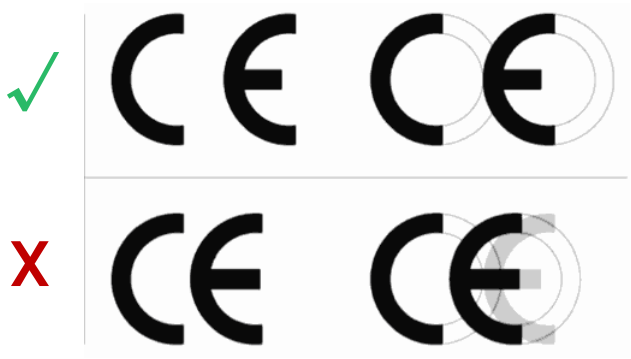
Machinery directive recital 2:
The machinery sector is an important part of the engineering industry and is one of the industrial mainstays of the Community economy. The social cost of the large number of accidents caused directly by the use of machinery can be reduced by inherently safe design and construction of machinery and by proper installation and maintenance.

Machinery directive recital 3:
Member States are responsible for ensuring the health and safety on their territory of persons, in particular of workers and consumers and, where appropriate, of domestic animals and goods, notably in relation to the risks arising out of the use of machinery.

Machinery directive article 6:
Member States shall not prohibit, restrict or impede the placing on the market and/or putting into service in their territory of machinery which complies with this Directive.

European Product Directives – CE-marking

- **C**onformité **E**uropéenne
- Guarantee of free trade within the EEA (European Economic Area)



Tab1-1/5-Sh.7

European Product Directives – CE-marking

- Safety and cost aspects
- Removing trade barriers

This is regulated by:

1. Harmonisation of laws (through Directives)
2. Normalisation (through European Standards)
3. Testing / certification (self-certification or Notified Body)

Tab1-1/5-Sh.8

Product Directives – CE marking

For example: Toy Safety Directive

(b) Toys and their parts must not present a risk of asphyxiation by closing off the flow of air as a result of airway obstruction external to the mouth and nose.

(c) Toys and their parts must be of such dimensions as to not present a risk of asphyxiation by closing off the flow of air as a result of internal airway obstruction by objects wedged in the mouth or pharynx or lodged over the entrance to the lower airways.

(d) Toys, which are clearly intended for use by children under 36 months, and their component parts and any of their detachable parts must be of such dimensions as to prevent their being swallowed or inhaled. This also applies to other toys which are intended to be put in the mouth, and to their component parts and any of their detachable parts.

Social directive– no CE marking

For example: display screen equipment

- Employers are required to carry out an analysis of workplaces in order to evaluate the safety and health conditions which they impose on their workers, in particular any risks to eyesight and problems of physical and mental strain.

Product and Social Directives

Product directives art. 114 (CE)	Social directives art. 153 (H&S)
• Machinery directive • EMC directive • Low voltage directive • Pressure equipment directive • ATEX directive • Gas appliances directive • Personal protective equipment directive	• Work equipment directive • Manual handling directive • Directive for safety and/or health signalling at work • Directive on temporary and mobile construction sites • Directive on working with display screen equipment

Sh.11

Product and Social Direct

Some examples of European Product Directives (CE)

- Machinery directive
- EMC directive
- Low voltage directive
- Pressure equipment directive
- ATEX directive

Included in the Dutch Commodities Act.

Tab1-1/5-Sh.12

CE-marking

European Union

Belgium	France	Lithuania	Romania
Bulgaria	Greece	Luxembourg	Slovenia
Cyprus	Hungary	Malta	Slovakia
Denmark	Ireland	Netherlands	Spain
Germany	Italy	Austria	Czech Republic
Estonia	Croatia	Poland	Finland
Latvia	Portugal	Sweden	

Meaning:

- Conformité Européenne
- Ensuring free trade in the EEA.
- EEA = European Economic Area = EU-states + EVA-states (Liechtenstein, Norway, and Iceland)

EU states which form part of the EEA
EFTA states which form part of the EEA
EU state which forms part of the EEA through the provisional application of an accession agreement
EFTA state which signed the EEA agreement but did not join
Former EU state which was part of the EEA

Tab1-1/5-Sh.13

Exporting outside the EER

When a machine is intended for a customer outside the EEA:

- It may still be brought into compliance with European product regulations and be provided with CE marking.
- The legal requirements applicable to deliver the machine in that country must be determined on a per-country basis.
- This may mean that (in addition to CE), other markings may be required to indicate compliance with the product:

Tab1-1/5-Sh.14

Exporting outside the EER

UK TIME FRAME GOODS for Lov
Machinery, pressure equipment, Radio Equ
Noise Emissions in th

Timeline to implement UKCA Marking

Present - 2022
You can use UKCA marking.

Until 30 June 2023
You can use CE marking on medical devices whilst you prepare to transition to UKCA.

Until 31 December 2024
You can use UKCA marking in most cases, whilst preparing to transition to UKCA marking.

After 31 December 2024
You must use UKCA marking when placing most manufactured products on GB market.

Until 31 December 2027

After 31 December 2027
UKCA marking must be applied directly on to the product unless legislations allow otherwise. Also, for ALL Importer marking

The Department for Business and Trade (DBT) has today announced its intention to indefinitely extend the use of CE marking for businesses, applying to 18 regulations owned by DBT.

This comes as part of a wider package of smarter regulations designed to ease business burdens and help grow the economy by cutting barriers and red tape. Following extensive engagement with industry, British firms will be able to continue the use of CE marking alongside UKCA.

The Business Secretary acted urgently on this issue, to prevent a cliff-edge moment in December 2024 when UKCA was set for entry. This intervention will ensure businesses no longer face uncertainty over the regulations and can cut back on unnecessary costs freeing them up to focus on innovation and growth.

Objectives European directives

Objectives product directives: (Manufacturer: CE-marking)

- Minimum requirements for products
- Reducing risks of products
- Removal of trade barriers (internal market)

Objectives of social directives: (Employer: health and safety at the workplace)

- Minimum requirements to improve health and safety in the workplace
- Harmonize national legislations (internal market)

Machine lifecycle

Case 1 : Completed machinery ready for transport:									
Design	Construction	Applying CE	Transport	Commissioning	Production	Maintenance	Inspection	Repairs	Dismantling
Responsibility: **Manufacturer**(CE-marking)				Responsibility: **Employer** (Health and safety employees)					
Case 2 : Machinery that is completed at the customer's location:									
Design	Construction	Transport	Complete construction and applying CE on-site	Commissioning	Production	Maintenance	Inspection	Repairs	Dismantling
Responsibility: **Manufacturer**(CE-marking)				Responsibility: **Employer** (Health and safety employees)					

The manufacturer must consider potential risks in the risk assessment that may occur during all phases of the machine's lifecycle (up to and including dismantling), starting from the CE marking.

Tab1-1/5-Sh.17

Machinery directive	Work equipment directive
• Product directive • Requires CE marking • Focused on manufacturer • Embedded in "de Warenwet (NL)" • Applies to machinery placed on the market since 1995	• Social directive • Does not require CE marking • Focused on employer • Embedded in "de Arbowet (NL)" • Applies to all work equipment (including machines) regardless of the year of manufacture
• Annex 1: Essential health and safety requirements concerning the design and construction of machinery	• Annex 1: Minimum requirements
• Requires that machinery is supplied with an instruction manual and a EC declaration of conformity	• Requires that the work equipment is provided with instructions for use
• Requires the manufacturer to compose a technical file	• Requires documenting a maintenance and inspection history

Tab1-1/5-Sh.18

General layout Product directive

Recitals

Articles

Annexes (selected)

1. Essential health and safety requirements
2. EC-Declaration of conformity
3. CE-marking
4. Annex IV machinery
5. List of safety components
6. Assembly instructions partly completed machinery
7. Technical file / Technical documentation

Tab1-1/5-Sh.19

Economic operators and responsibilities

Manufacturer:
When are you considered a manufacturer:

1. You manufacture or assemble a ('CE-mandatory') product yourself for sale or for your own use.
2. You modify a product or expand it, making the product considered 'as new.'
3. You sell a product under your own brand name (private label).
4. You import a non-CE-marked product from a non-EEA country.
5. You link various (new) individual machines into an assembly of machines (a production line).

See definition of assembly of machinery in module Tab1-2

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.20

Economic operators and responsibilities

Manufacturer's responsibilities:

- Responsible for designing and manufacturing the product in accordance with the requirements of the applicable EU legislation.
- Responsible for the conformity assessment and compilation of the required technical documentation.
- Preparing the EU Declaration of Conformity, instructions, and safety information and providing them with the product, as well as affixing the conformity markings (including the CE marking).
- Keeping the EU Declaration of Conformity and the technical documentation available.
- Providing documentation (to the authorities) to demonstrate product conformity and cooperating in taking corrective actions in case of risks and suspected non-compliance.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.21

Economic operators and responsibilities

Authorized Representative:

- The authorized representative is established in the EU ('authorized representative established in the community').
- Commercial representatives of the manufacturer (such as distributors or agents) should not be confused with the authorized representative.
- Regardless of whether the manufacturer is established in the EU or not, they may appoint an authorized representative in the Union to perform certain tasks required under the applicable Union harmonization legislation on their behalf. A manufacturer established outside the EU is not obliged to have an authorized representative.
- The tasks that may be delegated to the authorized representative under Union harmonization legislation are administrative in nature.
- The manufacturer may not delegate measures to ensure that the manufacturing process ensures that the products comply or the preparation of the technical documentation.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.22

Economic operators and responsibilities

Authorized Representative responsibilities :

If the manufacturer appoints an authorized representative, the authorized representative must be authorized (in writing) to perform the following tasks at least:

- Keeping the EU Declaration of Conformity and the technical documentation available.
- Providing documentation to demonstrate product conformity.
- Cooperating in corrective actions in case of risks and suspected non-compliance.

In addition, the following tasks may be entrusted:

- Applying conformity marking.
- Preparing EU Declaration of Conformity.
- Informing market surveillance authorities in case of suspected non-compliance.
- Verifying whether the EU Declaration of Conformity and technical documentation have been prepared.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.23

Economic operators and responsibilities

Importer:
The importer is a natural or legal person established in the Union who places a product from a third country on the market in the Union.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.24

Economic operators and responsibilities

Importer Responsibilities:

- Verifying that the manufacturer has carried out the appropriate conformity assessment procedure. In case of doubt, the importer may not place the product on the market. If already on the market, corrective measures must be taken. In these cases, contact must be made with the manufacturer to resolve any doubts about conformity.
- Verifying that the manufacturer has prepared the technical documentation, affixed the necessary conformity markings (e.g., CE marking), complied with traceability obligations, and included instructions and safety information with the product.
- Including the name, registered trade name or trademark, and postal address on the machine, packaging, or documentation.
- Keeping available the EU Declaration of Conformity and the technical documentation

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.25

Economic operators and responsibilities

Importer Responsibilities:

- Providing documentation to demonstrate product conformity.
- Cooperating in corrective measures in case of risks and suspected non-compliance.
- Identifying each economic operator who has supplied to him..

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.26

Economic operators and responsibilities

Distributor:

The distributor is a natural or legal person in the supply chain, different from the manufacturer and the importer, who offers a product on the market.

The distributor does not have a preferential relationship with the manufacturer and does not place products on the European market or put them into service.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Economic operators and responsibilities

Distributor responsibilities:

- Verifying the presence of required conformity markings.
- Verifying the presence of the EU Declaration of Conformity and user manual.
- Verifying the presence of the manufacturer's and importer's name, registered trade name or trademark, and contact address on the product, and that a type, batch, or serial number, or another means of identification, is provided.
- Keeping available the EU Declaration of Conformity.
- Identifying the economic operator who supplied to him and to whom he supplied the product for the purpose of market surveillance authorities..

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Economic operators and responsibilities

Distributor responsibilities:

- Providing documentation to demonstrate product conformity.
- Informing market surveillance authorities in case of suspected non-compliance.
- Cooperating in corrective measures in case of risks and suspected non-compliance.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.29

Economic operators and responsibilities

Fulfilment service providers:
They are natural persons or legal entities that, as part of a commercial activity, offer at least two of the following services: storage, packaging, addressing, and dispatching without being the owner of the products involved.

Fulfilment service providers are considered market participants within the meaning of Regulation (EU) 2019/1020 and must cooperate with market surveillance authorities regarding the products they handle.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Tab1-1/5-Sh.30

Economic operators and responsibilities

Fulfilment service providers:

- Verifying the presence of required conformity markings.
- Verifying the presence of the EU Declaration of Conformity and technical documentation.
- Keeping available the EU Declaration of Conformity and technical documentation.
- Providing documentation to demonstrate product conformity.
- Informing market surveillance authorities in case of suspected non-compliance.
- Initiating cooperation in corrective measures in case of risks and suspected non-compliance.

(See also Regulation on Market Surveillance 2019/1020 and the Blue Guide.)

Economic operators and responsibilities

Scenario:

A machine is ordered by an end user in the EU on April 1, 2023, based on an offer/model in a catalogue. The machine is then produced in China and shipped to the end user on June 1, 2023. It arrives at customs on June 20, 2023. The product is sold directly to the customer in the EU by the manufacturer outside the EU through distance selling.

Question 1-1: In this scenario, what is the date on which this machine is formally placed on the market?

Question 1-2: Can the manufacturer from outside the EU sell this machine directly to the end user located in the EU without the involvement of an authorized representative/importer or distributor?

Who applies CE-marking?

Quote from the 'Blue Guide 2022:

"The manufacturer may design and manufacture the product himself. As an alternative, he may have it designed, manufactured, assembled, packed, processed or labelled with a view to placing it on the market under his own name or trademark, and thus presenting himself as a manufacturer.

Where subcontracting takes place, the manufacturer must retain the overall control for the product and ensure that he receives all the information that is necessary to fulfil his responsibilities according to the relevant Union harmonisation act.

The manufacturer who subcontracts some or all of his activities may in no circumstances discharge himself from his responsibilities, for example to an authorised representative, a distributor, a user or a subcontractor."

Liability

Product liability:
- Product Liability Act (Civil Code)
- As of November 1, 1990

Burgerlijk wetboek:
- Productaansprakelijkheid: de producent (fabrikant) is aansprakelijk voor de schade veroorzaakt door een gebrek in zijn product.
- Unlawful act: He who commits an unlawful act against another, which can be attributed to him, is obliged to compensate the damage suffered by the other as a result.
- In the case of an accident involving machinery where the manufacturer is accused of placing an unsafe product on the market, the manufacturer must be able to demonstrate with the Technical Documentation that there is no causal link between the incurred injury and an unsafety in the product (/the machinery).

Rapex (notification by competent authorities):
Safety Gate: the EU rapid alert system for dangerous non-food products
https://ec.europa.eu/safety-gate-alerts/screen/search?resetSearch=true

Arbeidsinspectie (zelf melden):
https://portaal.nlarbeidsinspectie.nl/productveiligheid

Question 1-3

If an end user (production company) purchases various new machinery from third parties and has them assembled by their own technical service into an assembly of machinery, does the end user then need to apply CE marking themselves and thereby comply with all obligations (essential requirements) of the applicable product directives?

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